

MATHEMATICS

Max Marks: 20

TIME: 1hr

Instructions:

- All questions are compulsory.
- Answer all sections in the given sequence.
- Show all necessary working for calculations.

Section A:

Very Short Answer (1 marks each, 4 marks total)

Q.1 Suhana sells a sofa set for Rs 9600 making a profit of 20%. What is the C.P. of the sofa set?

Q.2 Solve :

- (a) 20% of 700 m is
- (b) 12% of 120

Q.3 Find the value of x :

- (a) 45% of marks x is 405
- (b) 32% of x kg is 400 kg

Q.4 State True or False:

- (a) The value of a variable is fixed
- (b) Every polynomial is a monomial

Section B:
Short Answer (2 marks each, 16 marks total)

Q.1 Subtract the sum of $-3x^3y^2 + 2x^2y^3$ and $-3x^2y^3 - 5y^4$ from $x^4 + x^3y^2 + x^2y^3 + y^4$

Q.2 Simplify the expression by combining the like terms:

(a) $7x^3 - 3x^2y + xy^2 + x^2y - y^3$

Q.3 What should be added to $x^3 + 3x^2y + 3xy^2 + y^3$ to get $x^3 + y^3$

Q.4 Solve :

(a) Medha deposited 20% of her money in a bank. After spending 20% of the remainder, she has Rs 4800 left with her. How much did she originally have?

(b) The cost of a flower vase got increased by 12%. If the current cost is Rs 896, what was its original cost?

Q.5 Solve :

(a) 8 hours is 50% of 4 days.

(b) What sum of money lent out at 16 per cent per annum simple interest would produce 9600 as interest in 2 years?

Q.6 Subtract

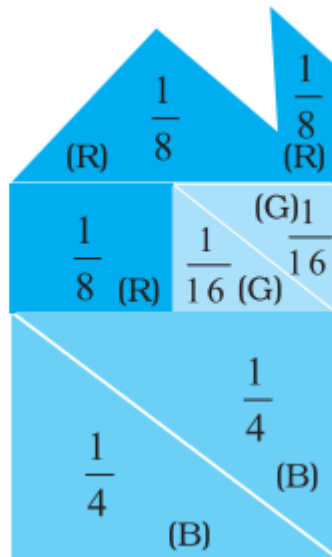
(a) $-2a^2 - 2b^2$ from $-a^2 - b^2 + 2ab$

(b) $x^3y^2 + 3x^2y^2 - 7xy^3$ from $x^4 + y^4 + 3x^2y^2 - xy^3$

Q.7 What should be added to $3pq + 5p^2q^2 + p^3$ to get $p^3 + 2p^2q^2 + 4pq$

Q.8 Solve :

(a) The pieces of Tangrams have been rearranged to make the given shape:



(b) Amount received after depositing Rs.800 for a period of 3 years at the rate of 12% per annum is Rs.896.